INCH-POUND
MIL-PRF-3098/52G
9 December 2004
SUPERSEDING
MIL-PRF-3098/52F
27 August 1997

PERFORMANCE SPECIFICATION SHEET

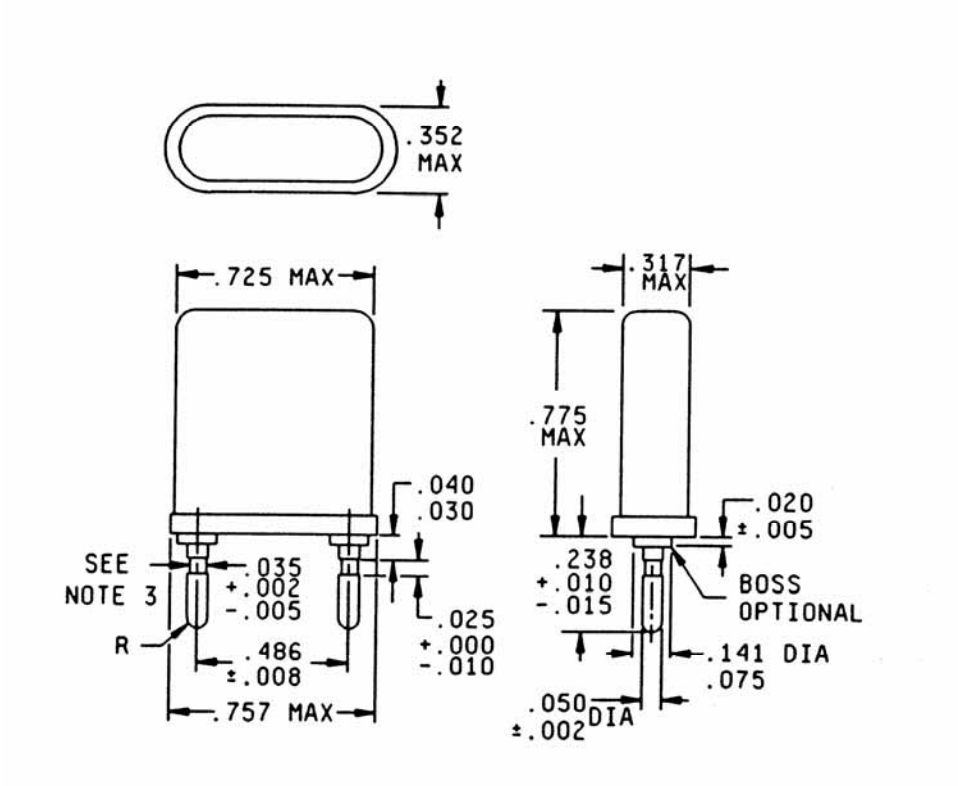
CRYSTAL UNIT, QUARTZ, CR75/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 50 MHz to 125 MHz; fifth mechanical overtone; controlled; series resonance.

Inches	mm
.002	0.05
.005	0.13
.008	0.20
.010	0.25
.015	0.38
.020	0.51
.025	0.64
.030	0.76
.035	0.89
.040	1.02
.050	1.27
.075	1.91
.141	3.58
.238	6.05
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
.775	19.69



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR75/U.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:32Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 50.0 MHz to 125 MHz, inclusive.

Frequency tolerance:

Operating temperature range: ±10 parts per million (ppm).

Room temperature: ±70 ppm.

Equivalent resistance: ±40 ohms, maximum.

Mode of oscillation: Fifth mechanical overtone.

Reference temperature: +75°C ±1°C.

Temperature ranges:

Operable temperature range: -55°C to +70°C and +80°C to +90°C, inclusive.

Operating temperature range (controlled): +70°C to +80°C, inclusive.

Drive level: 1.0 mW maximum.

Resonance: Series.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:32Z
Check the source to verify that this is the current version before use.

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC, SH
Air Force – 19

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0781)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:32Z
Check the source to verify that this is the current version before use.